# Robotics Assignment

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## 1. Understand the concept of homogeneous transformation matrices

### Introduction

In robotics, a robot manipulator consists of several rigid links connected by joints. Each link has its own coordinate frame. To determine the position and orientation of one frame relative to another, a homogeneous transformation matrix (HTM) is used.

A homogeneous transformation matrix combines rotation and translation into a single matrix operation. Instead of applying rotation and translation as two separate operations, homogeneous coordinates represent both using one matrix multiplication.

### Why Homogeneous Coordinates?

A point in Cartesian coordinates is represented in 2D as:

$$P = \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation can be written as matrix multiplication $P' = RP$, where:

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

However, translation cannot be represented by matrix multiplication alone. Translation requires addition:

$$P' = \begin{bmatrix} x + t_x \\ y + t_y \end{bmatrix}$$

To express translation as a matrix multiplication, one extra coordinate is added. A 2D point becomes:

$$P = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

A 3D point becomes:

$$P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

## General Homogeneous Transformation Matrix

In 2D, the transformation matrix $T$ is defined as:

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$$

where the rotation matrix $R$ and translation vector $t$ are:

$$R = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}, \qquad t = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Expanding this gives:

$$T = \begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

In 3D, the transformation matrix $T$ follows the same structure:

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$$

where $R$ is a $3 \times 3$ matrix and $t$ is a $3 \times 1$ vector:

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}, \qquad t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Thus, the full 3D transformation matrix is:

$$T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

## Components and Properties

**Rotation Matrix ($R$):** Represents orientation.

- $R^T R = I$
- $R^{-1} = R^T$
- $\det(R) = 1$

**Translation Vector ($t$):** Represents displacement of the frame origin.

$$t = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

**Transformation of a Point:**

If $P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$, then the transformed point is calculated as:

$$P' = TP$$

which is mathematically equivalent to:

$$P' = RP + t$$

### Importance in Robotics

Homogeneous transformations are fundamental because they:

- Represent both position and orientation simultaneously.
- Simplify forward kinematics.
- Enable chaining of multiple coordinate transformations via matrix multiplication.
- Are essential for Denavit–Hartenberg (DH) parameter modeling.

## 2. Perform translation in planar (2D) and spatial (3D) coordinate systems

### Definition

Translation moves an object from one location to another without changing its orientation or shape. If the translation vector is $t = (t_x, t_y)$, then the new coordinates are:

$$x' = x + t_x$$
$$y' = y + t_y$$

### Translation in 2D

The homogeneous translation matrix is:

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

For a point $P = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, the transformed point becomes:

$$P' = TP$$

**Example**

Translate the point $(2, 3)$ by the vector $(5, -1)$.

**Matrix:**

$$T = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

**Point:**

$$P = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

**Result:**

$$P' = TP = \begin{bmatrix} 1(2) + 0(3) + 5(1) \\ 0(2) + 1(3) - 1(1) \\ 0(2) + 0(3) + 1(1) \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 1 \end{bmatrix}$$

### Translation in 3D

The homogeneous translation matrix is:

$$T = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

**Example**

Translate the point $(1, 2, 3)$ by $(4, -2, 5)$.

The result is simply the vector addition:

$$(1 + 4, 2 - 2, 3 + 5) = (5, 0, 8)$$

### Properties

- Shape remains unchanged.
- Orientation remains unchanged.
- Distances between points remain constant.
- Parallel lines remain parallel.

## 3. Perform rotation in planar and spatial coordinate systems

### Definition

Rotation changes the orientation of an object about a fixed axis while preserving its size and shape.

## Rotation in 2D

Rotation by angle $\theta$ about the origin is given by the rotation matrix:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

The homogeneous form of this rotation is:

$$T = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

### Example

Rotate the point $(1, 0)$ by $90°$.

Since $\cos(90°) = 0$ and $\sin(90°) = 1$, the transformation matrix becomes:

$$T = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Applying this to the homogeneous point:

$$P' = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

**Result:** The transformed point is $(0, 1)$.

## Rotation in 3D

In spatial coordinate systems, three principal axes are used.

**Rotation about the X-axis:**

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

**Rotation about the Y-axis:**

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

**Rotation about the Z-axis:**

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

### Homogeneous Form

For rotation about the Z-axis, the $4 \times 4$ homogeneous transformation matrix is:

$$T = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

*(The homogeneous forms for X and Y axis rotations follow the same $4 \times 4$ expansion pattern.)*

## Properties

- Preserves length of vectors.
- Preserves angles between vectors.
- Rotation matrices are orthogonal, meaning $R^T R = I$ and $R^{-1} = R^T$.
- Multiple rotations are combined using matrix multiplication.

# 4. Combine translation and rotation into a single homogeneous transformation

A robot rarely performs only translation or only rotation. Most robotic motions involve both simultaneously.

The combined homogeneous transformation matrix represents both operations in a single matrix:

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$$

where $R$ is the rotation matrix and $t$ is the translation vector.

## 2D Example

Rotate a frame by $45°$ and translate it by $(2, 3)$.

**Rotation Matrix:**

$$R = \begin{bmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{bmatrix}$$

**Combined Matrix:**

$$T = \begin{bmatrix} \cos 45° & -\sin 45° & 2 \\ \sin 45° & \cos 45° & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Applying this single matrix simultaneously rotates and translates the point.

### 3D Example

If the rotation is about the Z-axis by 30° ($R = R_z(30°)$) and the translation vector is $t = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$, then the combined transformation matrix is:

$$T = \begin{bmatrix} \cos 30° & -\sin 30° & 0 & 5 \\ \sin 30° & \cos 30° & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

### Advantages

- **Single matrix multiplication:** Both operations are computed in one step.
- **Simplifies forward kinematics:** Essential for calculating the end-effector position of a robotic arm.
- **Easy composition:** Multiple sequential transformations can be chained together simply by multiplying their respective homogeneous matrices.
- **Efficient implementation:** Highly optimized in modern robotics software and hardware architectures.

## 5. Compare different orders of transformation

Matrix multiplication is not commutative, meaning:

$$AB \neq BA$$

Therefore, the order in which translation and rotation are applied significantly affects the final position and orientation.

### Case 1: Rotation Followed by Translation

The transformation is applied by first rotating the point about the origin and then translating it. The combined matrix is:

$$T = T_t T_r$$

For a point $P$, the transformed position is:

$$P' = T_t T_r P$$

### Case 2: Translation Followed by Rotation

The transformation is applied by first translating the point and then rotating it about the global origin. The combined matrix is:

$$T = T_r T_t$$

For a point $P$, the transformed position is:

$$P' = T_r T_t P$$

### Example

Let the initial point be $P = (1, 0)$. Apply a translation of $(2, 0)$ and a rotation of $90°$.

**Rotation $\rightarrow$ Translation:**

- **Rotate:** $(1, 0) \rightarrow (0, 1)$
- **Translate:** $(0, 1) \rightarrow (2, 1)$

**Final point:** $(2, 1)$

**Translation $\rightarrow$ Rotation:**

- **Translate:** $(1, 0) \rightarrow (3, 0)$
- **Rotate:** $(3, 0) \rightarrow (0, 3)$

**Final point:** $(0, 3)$

The final coordinates are different ($(2, 1)$ vs $(0, 3)$), demonstrating that transformation order mathematically matters.

### Comparison Table

| **Aspect** | **Rotation $\rightarrow$ Translation** | **Translation $\rightarrow$ Rotation** |
|---|---|---|
| **First operation** | Rotation | Translation |
| **Second operation** | Translation | Rotation |
| **Rotation center** | Original origin | Shifted point after translation |
| **Matrix form** | $T_t T_r$ | $T_r T_t$ |
| **Commutative?** | No | No |

### Conclusion

Homogeneous transformation matrices provide a unified mathematical framework for representing rotations and translations in robotics. They simplify coordinate transformations, enable efficient computation of robot kinematics, and allow multiple transformations to be combined through matrix multiplication. Since transformation matrices are not commutative, the sequence in which rotations and translations are applied must be chosen carefully, as it directly influences the final pose of the robot or object.

## 6. Visualize transformed coordinate frames using Python Robotics Toolbox

### Practical Implementation

Visualization of coordinate frame transformations is essential for verifying theoretical calculations in robotics. The Python Robotics Toolbox (`roboticstoolbox-python`) provides robust functions to plot 2D and 3D coordinate frames after applying translation, rotation, and combined homogeneous transformations.

The computational implementation, including the generation of transformation matrices and the resulting spatial plots, has been executed and documented in a Kaggle notebook.

https://www.kaggle.com/code/kalpanabhaskar/robotics-spatial-transformation
*Note: Open the link above to view the Python source code and the interactive coordinate frame visualizations.*